AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 26 · STUDENT LESSON

Inspecting failures and anomalies

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Running the primary experiment. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of inspecting failures and anomalies. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of inspecting failures and anomalies.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of inspecting failures and anomalies

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Running the primary experiment. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain inspecting failures and anomalies using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Inspecting failures and anomalies case. Separate observations, given facts and assumptions before choosing an explanation.

Failure inspection generates explanations to test. An unusual result is not automatically a bad data point.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Inspecting failures and anomalies. Connect each labelled input, deciding step and result.

Compare raw **output**s, **input** validity and procedure traces. Correct demonstrable measurement errors with provenance; retain valid unusual **observation**s. Label post-result investigations exploratory.

Inspecting failures and anomalies — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Durations1, 2, 2, 15 seconds are all valid traces. Team deletes15 as an outlier without a rule.	Compare raw outputs, input validity and procedure traces. Correct demonstrable measurement errors with provenance; retain valid unusual observations.	The slow valid case must remain in the declared primary analysis. Investigate mechanism separately; mean5 and median2 reveal different aspects.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Inspecting failures and anomalies. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Durations1, 2, 2, 15 seconds are all valid traces. Team deletes15 as an outlier without a rule. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Delete the 10-second record only because it looks bad.

Predicted result

Prediction: The mean becomes 2 seconds but the selective report violates this plan.

Why the result follows

Reason: Observed inconvenient outcomes are not an approved exclusion reason.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Inspecting failures and anomalies to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Anomaly	An unusual observation requiring investigation
Provenance-backed correction	A repair justified by source evidence
Exploratory analysis	A follow-up chosen after results

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Inspecting failures and anomalies, predict a step, then compare with its explanation.

Durations1, 2, 2, 15 seconds are all valid traces. Team deletes15 as an outlier without a rule.

The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean5 and median2 reveal different aspects.

Worked explanation · Inspecting failures and anomalies

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Inspecting failures and anomalies

Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration.

The facts we will use

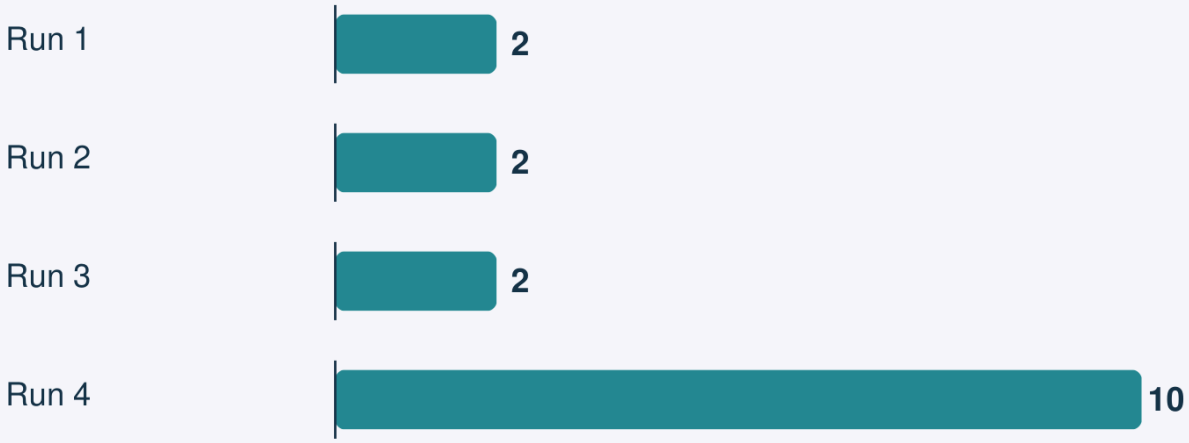
Supplied facts

Durations are 2,2,2,10 seconds. Logs show a source timeout on the fourth run. The plan includes timeouts in reporting.

AI ACADEMY / AI-7/26

All four recorded durations

seconds; run 4 includes timeout



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/26

Inspecting failures and anomalies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep the raw record

The fourth duration remains 10 seconds.



2. Compute summaries

Mean is 16/4=4 seconds; median is 2 seconds.



3. Investigate the event

Report the timeout evidence and distinguish it from an untested explanation.

CHANGE ONE THING

Delete the 10-second record only because it looks bad.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep the raw record	The fourth duration remains 10 seconds.
2. Compute summaries	Mean is 16/4=4 seconds; median is 2 seconds.
3. Investigate the event	Report the timeout evidence and distinguish it from an untested explanation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

An **anomaly** invites investigation, not automatic deletion. A slow failure may be part of real service behaviour; the predefined rule determines inclusion.

What this example does not establish

Keep the stated limit

A logged timeout does not identify every underlying cause; report conditions and test explanations.

Keep the stated limit

Read the labelled picture: Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?

Keep the stated limit

Change one thing: Delete the 10-second record only because it looks bad. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Inspecting failures and anomalies claim and explain a repair using the supplied evidence.

Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Post-hoc exclusion based on an unwanted outcome can distort the conclusion. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Inspecting failures and anomalies with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Inspecting failures and anomalies”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Inspecting failures and anomalies. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Inspecting failures and anomalies. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Durations1, 2, 2, 15 seconds are all valid traces. Team deletes15 as an outlier without a rule. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Inspecting failures and anomalies. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: Post-hoc exclusion based on an unwanted outcome can distort the conclusion. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled inspecting failures and anomalies **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of inspecting failures and anomalies.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Inspecting failures and anomalies case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Inspecting failures and anomalies — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Durations1, 2, 2, 15 seconds are all valid traces. Team deletes15 as an outlier without a rule.	Compare raw outputs, input validity and procedure traces. Correct demonstrable measurement errors with provenance; retain valid unusual observations.	The slow valid case must remain in the declared primary analysis. Investigate mechanism separately; mean5 and median2 reveal different aspects.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Values2, 3, 4, 20 are valid. Compute mean and median and propose an honest investigation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Values2, 3, 4, 20 are valid. Compute mean and median and propose an honest investigation.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Inspecting failures and anomalies, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration.

By the end, I can	How I will show it
Explain the mechanism and terminology of inspecting failures and anomalies	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Inspecting failures and anomalies without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of inspecting failures and anomalies

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report the timeout **evidence** and distinguish it from an untested explanation.

1. Investigate the event

2. Keep the raw record

3. Compute summaries

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Investigate the event Report the timeout **evidence** and distinguish it from an untested explanation. An **anomaly** invites investigation, not automatic deletion.

A slow failure may be part of real service behaviour; the predefined rule determines inclusion.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Delete the 10-second record only because it looks bad. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The mean becomes 2 seconds but the selective report violates this plan.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The mean becomes 2 seconds but the selective report violates this plan. Observed inconvenient outcomes are not an approved exclusion reason.

Repair a missing building block

Review Running the primary experiment (ai-7/25). Goal: Explain the **mechanism** and terminology of running the primary experiment. Short practice: Compute A's and B's six-case counts and identify the cases that favour each.

How must a B revision tuned on case four be labelled when evaluated again on the same set?. Return to this week and explain what you changed.

Review Power and resource tradeoffs (ai-7/23). Goal: Explain the **mechanism** and terminology of power and resource tradeoffs. Short practice: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish power for an actual classroom study?. Return to this week and explain what you changed.

Review Privacy and data governance (ai-7/20). Goal: Explain the **mechanism** and terminology of privacy and data governance. Short practice: Point to the analysis table and the restricted identity key.

Why are the codes pseudonymous rather than guaranteed anonymous, and what changes if the key is published?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

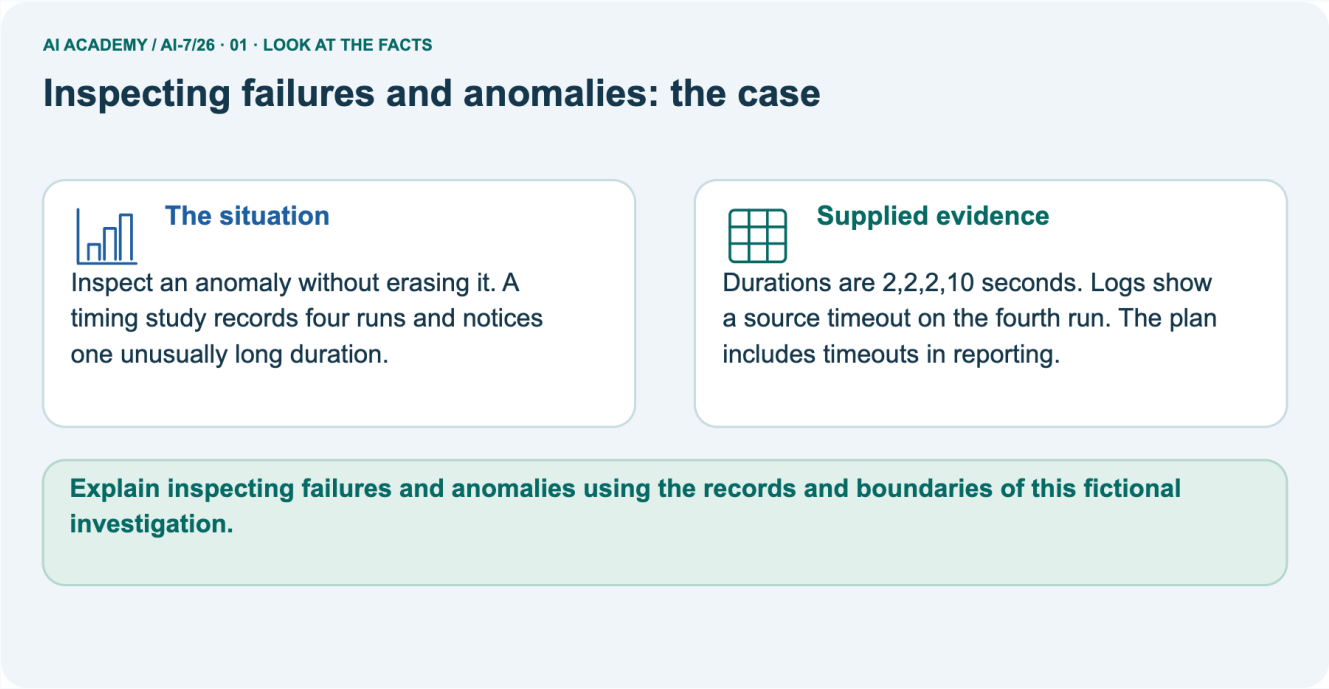
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Inspecting failures and anomalies graphic using its labels, deciding evidence and relevant limitation.

01. Inspecting failures and anomalies: the case



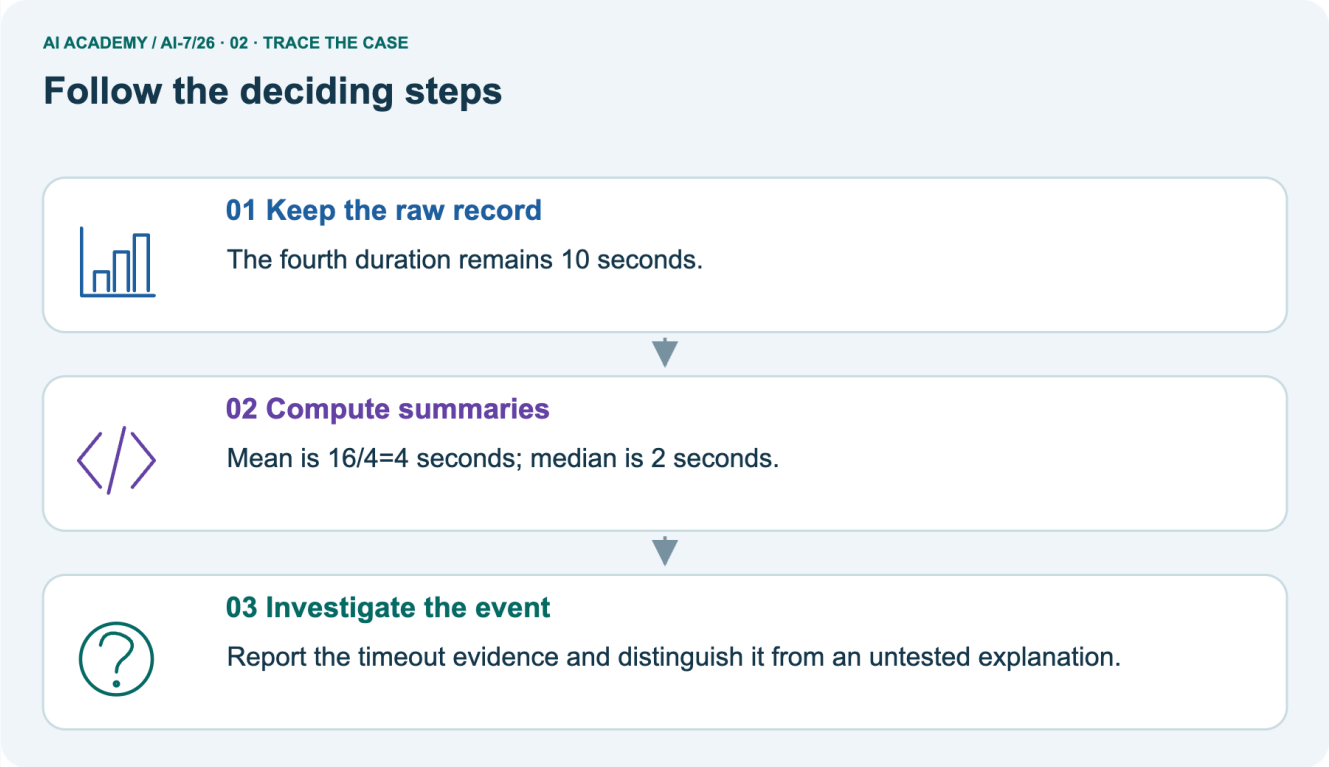
Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration. Supplied facts: Durations are 2,2,2,10 seconds. Logs show a source timeout on the fourth run.

The plan includes timeouts in reporting.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

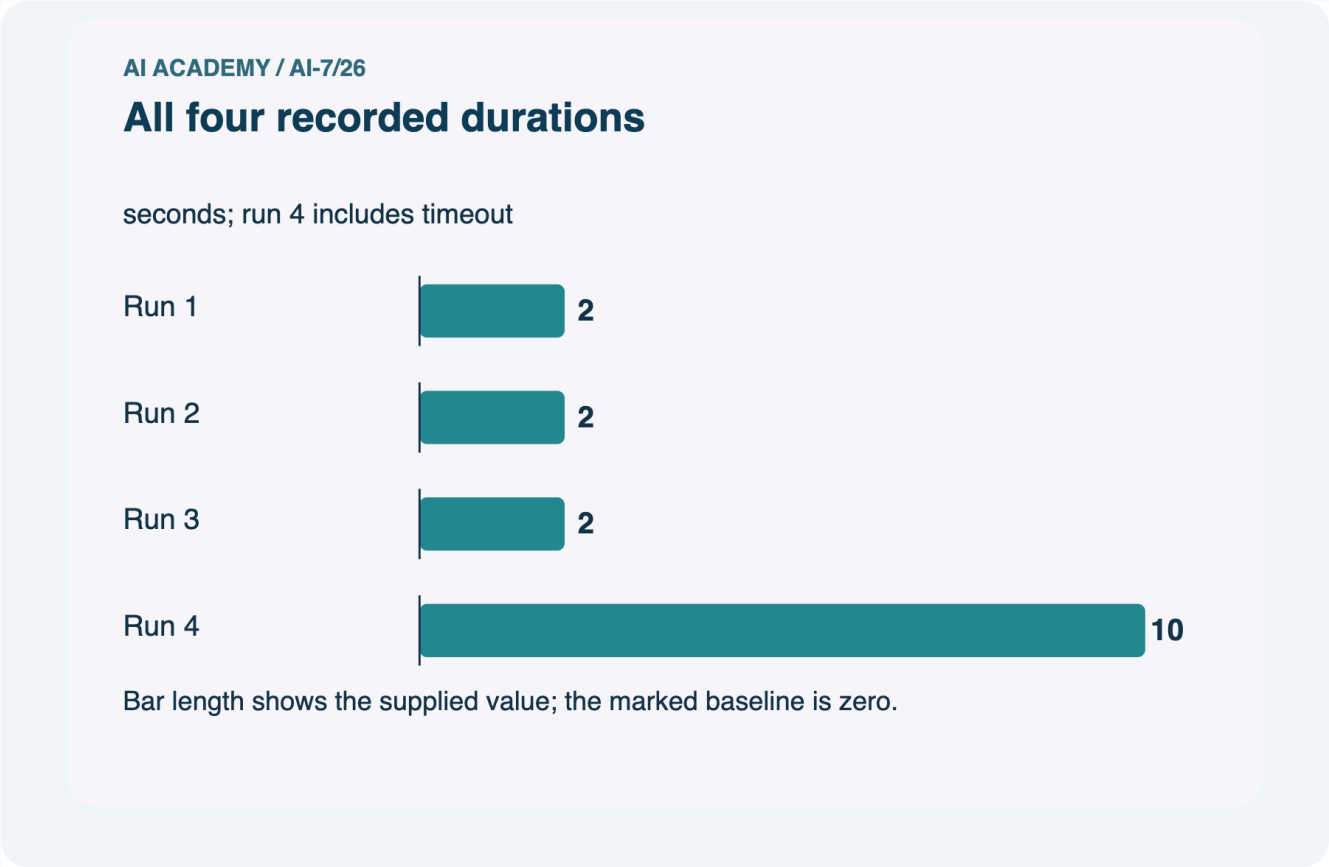


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the raw record: The fourth duration remains 10 seconds. → Compute summaries: Mean is 16/4=4 seconds; median is 2 seconds. → Investigate the event: Report the timeout **evidence** and distinguish it from an untested explanation.

Point to each numbered stage and explain which recorded **input** it uses.

03. All four recorded durations

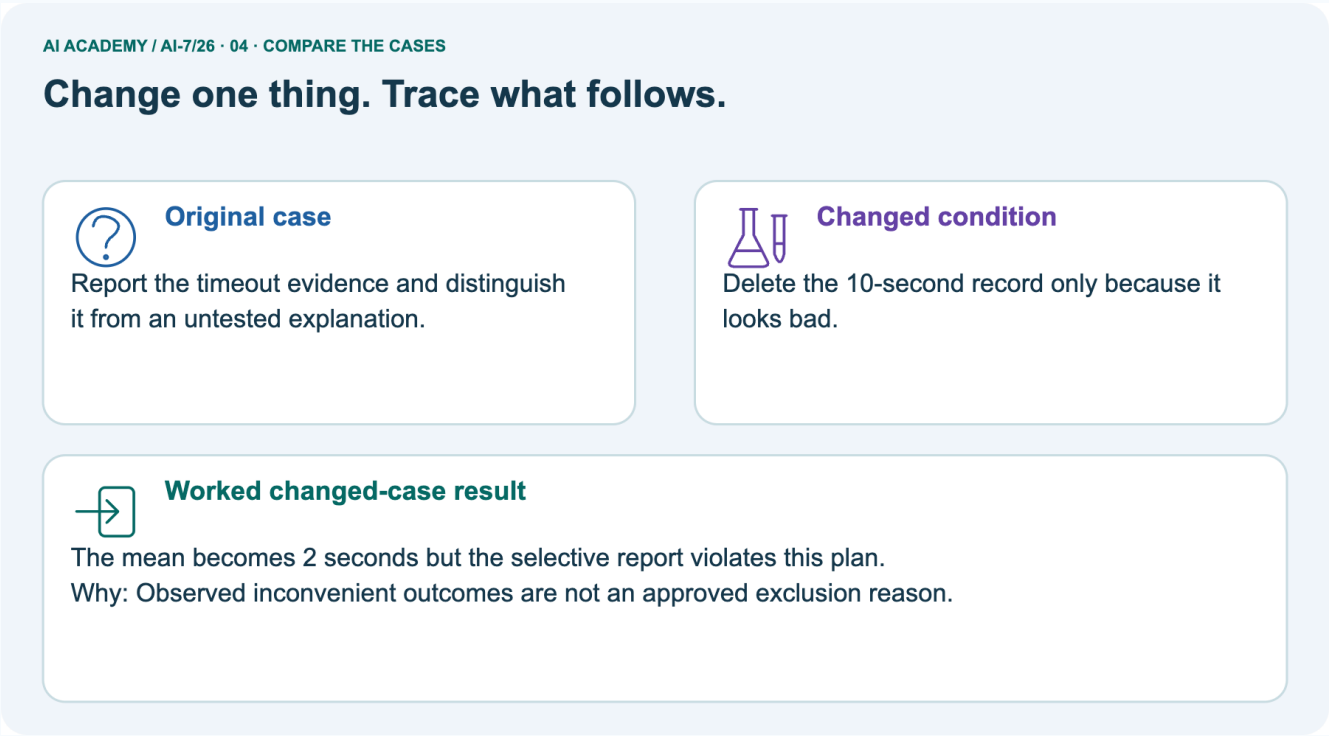


Follow the labelled picture with the matching explanation. Open the vector diagram

All four recorded durations. Units: seconds; run 4 includes timeout. Run 1: 2. Run 2: 2. Run 3: 2. Run 4: 10.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



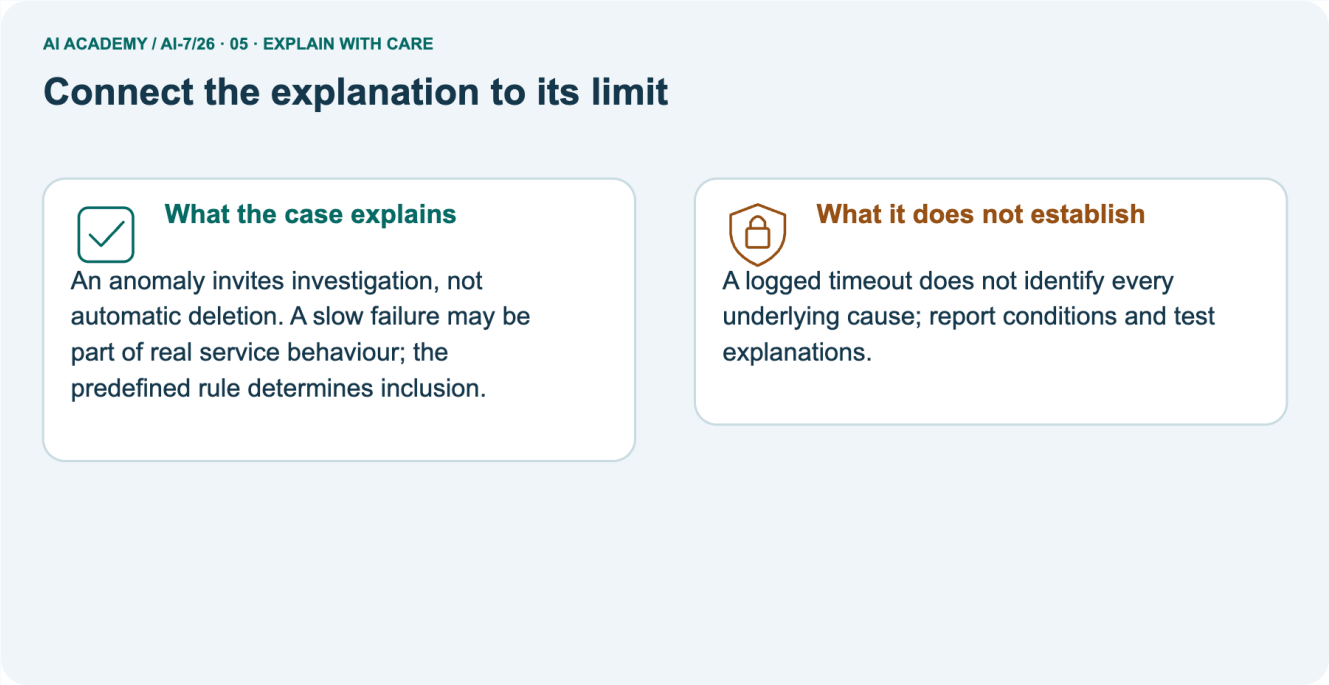
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report the timeout **evidence** and distinguish it from an untested explanation.
Changed condition: Delete the 10-second record only because it looks bad.

Expected result: The mean becomes 2 seconds but the selective report violates this plan. Reason: Observed inconvenient outcomes are not an approved exclusion reason.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



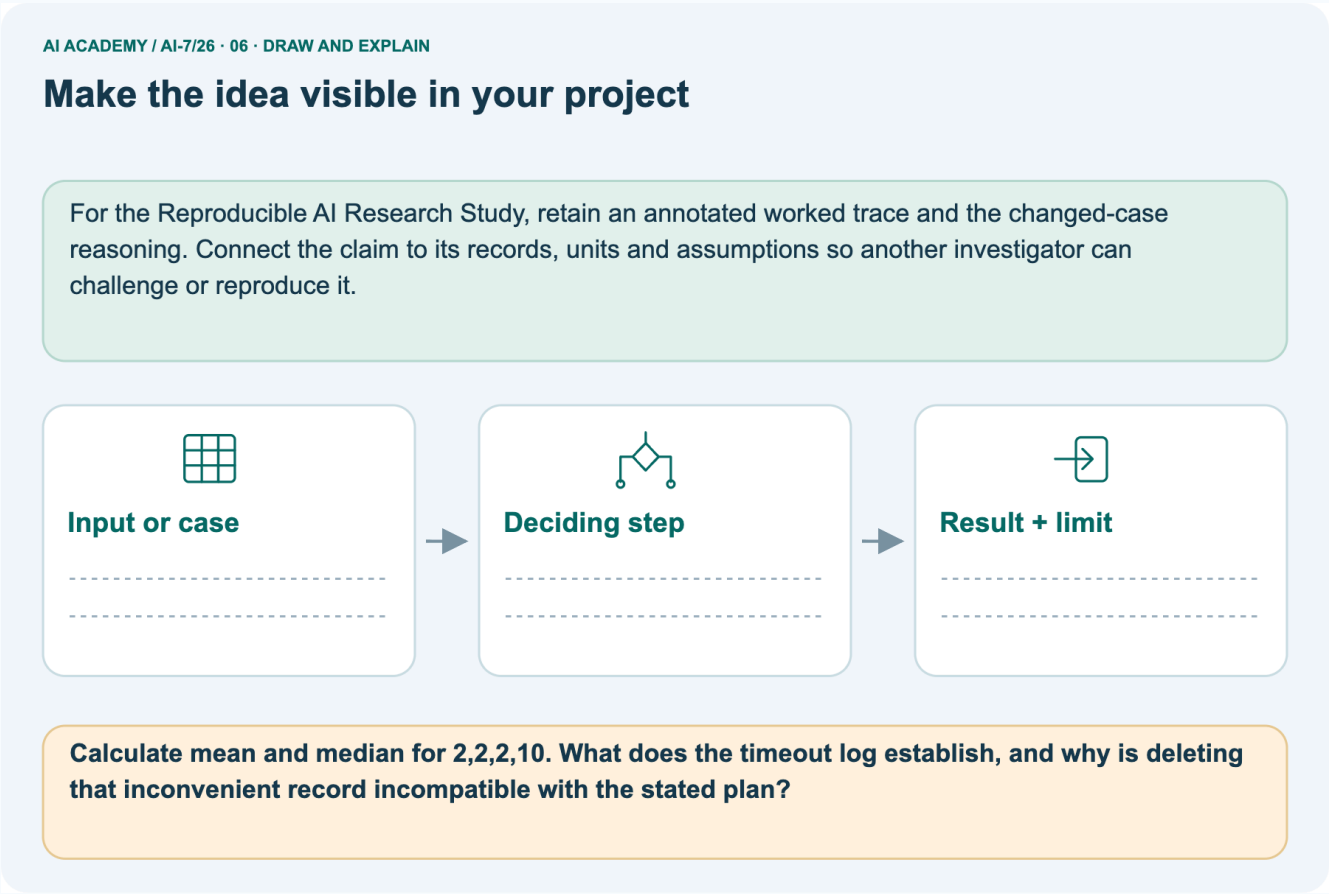
Follow the labelled picture with the matching explanation. Open the vector diagram

An **anomaly** invites investigation, not automatic deletion. A slow failure may be part of real service behaviour; the predefined rule determines inclusion.

Limit: A logged timeout does not identify every underlying cause; report conditions and test explanations.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.